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An LP-based heuristic for two-stage capacitated facility location problems

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In this paper, a linear programming based heuristic is considered for a two-stage capacitated facility location problem with single source constraints. The problem is to find the optimal locations of depots from a set of possible depot sites in order to serve customers with a given demand, the optimal assignments of customers to depots and the optimal product flow from plants to depots. Good lower and upper bounds can be obtained for this problem in short computation times by adopting a linear programming approach. To this end, the LP formulation is iteratively refined using valid inequalities and facets which have been described in the literature for various relaxations of the problem. After each reoptimisation step, that is the recalculation of the LP solution after the addition of valid inequalities, feasible solutions are obtained from the current LP solution by applying simple heuristics. The results of extensive computational experiments are given.

Keywords: LP-based heuristics; cutting planes; facility location

Introduction

Many distribution systems have a two-level structure. Central facilities (plants and central warehouses) supply intermediary facilities (regional depots, distribution centers and transshipment points) where the product flow is broken into smaller consignments which are directly shipped to the customers. In designing or redesigning the distribution system, one has to decide on the number and the location of the intermediary facilities. This decision can be supported by solving a two-stage capacitated facility location problem (TSCFLP). The problem consists in finding the optimal locations of depots to serve customers with a given demand, in order to arrive at an optimal assignment of customers to depots and to determine the product flow from the plants to the depots. The total cost consists of the cost of shipping the commodities from the plants to the depots, the cost of serving the customers from the depots, the cost of throughput at each depot and the fixed cost of maintaining the depots.

While there is a large body of literature on one-stage facility location problems, the work which has been carried out on two-stage or multi-stage location problems seems to be very limited. Kaufman *et al*¹ consider a two-level uncapacitated facility location problem (TUFL) with the additional constraint that there must be a depot at each plant. Problem instances with up to 5 plants, 30 depot sites and 50 customers are solved by a branch and bound algorithm which extends Efronson and Ray's² procedure for solving

the simple plant location problem (SPLP). Tcha and Lee³ extend Erlenkotter's⁴ dual ascent and adjustment procedure for the SPLP to a branch and bound method for multi-stage uncapacitated facility location problems. They reported on computational results for two-level problems involving 50 customers, 5 plants and 25 depot sites, and three-level problems with 50 customers and 8, 6 and 4 possible facility sites at each echelon. Gao and Robinson^{5,6} also extend Erlenkotter's dual-based algorithm to two-level and multi-commodity uncapacitated facility location problems. The largest problems they solve have 25 possible facility sites at each echelon and 35 customer zones. Aardal *et al*⁷ investigated the polyhedral structure of the TUFL and propose new valid inequalities and facets which can be used in branch and cut algorithms. Aardal^{8,9} and Aardal *et al*¹⁰ also study the cutting plane approach for one-level and two-level capacitated facility location problems. Aardal^{8,9} reported on the solution of one-level problems with up to 100 customers and 75 potential sites, and two-level problems with 50 customers, 16 possible depot sites and 6 possible plant sites using a cutting plane approach. The problem of optimally locating depots in a two-stage multicommodity distribution system is considered by Geoffrion and Graves.¹¹ They successfully apply the method of Benders decomposition to solve a real-world problem with 17 commodity classes, 14 plants, 45 possible depot sites and 121 customer zones. A similar distribution system design problem is addressed by Hindi and Basta;¹² however, in contrast to the model of Geoffrion and Graves, the customers need not be supplied by a single depot. Hindi and Basta used a branch and bound algorithm which is based on a weak LP relaxation of the problem and extends the procedure of

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Akinc and Khumawala¹³ for the one-level capacitated facility location problem (CFLP). The largest problem instances they solve have 3 products, 10 plants, 15 possible depot sites and 30 customers.

Real distribution system design problems often involve 50–200 potential facility sites and, depending on the degree of aggregation, several thousand customers.^{14–16} At the current state of the art in discrete location theory and mixed integer programming, it is not generally possible to obtain optimal solutions to problems of this size, especially if they contain single-source constraints. Therefore, it is important to have methods available which are able to produce good lower and upper bounds on the optimal objective value in a reasonable amount of computation time. In this paper, a linear programming based heuristic is described which computes such bounds for a capacitated two-stage facility location problem with single source constraints. Firstly, in the next section, the problem is formulated as a mixed integer programming problem. Secondly, some valid inequalities are summarised, which can be used to strengthen the LP relaxation and which have been described in the literature for various relaxations of this problem. Thirdly, simple heuristics are described for constructing feasible solutions from the current LP solution. Finally, after summarising the procedure, the results of extensive computational experiments are presented.

Problem formulation

The location problem considered in this study extends the CFLP with single sourcing (see Sridharan¹⁷) and is a one-product version of the distribution system design problem described by Geoffrion and Graves.¹¹ The problem can be formulated as the following mixed integer program:

$$\min \sum_{i \in I} \sum_{j \in J} t_{ij} v_{ij} + \sum_{k \in K} \sum_{j \in J} c_{kj} z_{kj} + \sum_{j \in J} f_j y_j \quad (1)$$

subject to

$$\sum_{j \in J} z_{kj} = 1 \quad \forall k \in K, \quad (2)$$

$$\sum_{k \in K} d_k z_{kj} \leq s_j y_j \quad \forall j \in J, \quad (3)$$

$$z_{kj} - y_j \leq 0 \quad \forall k \in K, j \in J, \quad (4)$$

$$\sum_{j \in J} s_j y_j \geq d(K), \quad (5)$$

$$\sum_{j \in J} v_{ij} \leq p_i \quad \forall i \in I, \quad (6)$$

$$\sum_{i \in I} v_{ij} = \sum_{k \in K} d_k z_{kj} \quad \forall j \in J, \quad (7)$$

$$v_{ij} - p_i y_j \leq 0 \quad \forall i \in I, j \in J, \quad (8)$$

$$v_{ij} \geq 0 \quad \forall i \in I, j \in J, \quad (9)$$

$$z_{kj} \in \{0, 1\} \quad \forall k \in K, j \in J, \quad (10)$$

$$y_j \in \{0, 1\} \quad \forall j \in J. \quad (11)$$

In this formulation, I denotes the set of plants, J the set of possible depot sites and K the set of customers; p_i is the production capacity of plant i , s_j the capacity of a depot at site j , d_k the demand of customer k , and $d(K)$ the total demand of a set K of customers. The cost of shipping one unit from plant i to depot j is denoted by t_{ij} , c_{kj} denotes the cost of supplying customer k from depot j , and f_j is the fixed cost of operating depot j . The variable y_j equals 1 if depot j is open and 0 otherwise, z_{kj} equals 1 if customer k is assigned to depot j and 0 otherwise, and v_{ij} is the flow from plant i to depot j . The constraints (2) are the demand constraints. The constraints (3) force z_{kj} to be 0 for all k if $y_j = 0$, and limit the throughput at each depot j to be not greater than its capacity s_j . The constraints (6) are the supply constraints and (7) are the ‘flow conservation constraints’. The constraint (5) is redundant in the MIP formulation and the LP relaxation as well. However, this surrogate constraint is helpful to derive additional cover inequalities which may strengthen the LP relaxation. The additional constraints (4) and (8) are (trivial) clique constraints and variable upper bounds respectively. They are redundant for the integer program, but strengthen the LP formulation. If the depot capacity (3) and supply constraints (6) are not too slack, only a small percentage of these constraints will be active in the optimal LP solution. The constraints (4) and (8) are therefore first omitted and only generated if violated by the current LP solution. The locational decisions refer only to the intermediary facilities at the second echelon. The two-stage nature of the problem thus vanishes if the supply constraints (6) are redundant. In this case, the problem can be reduced to a CFLP with single sourcing. If in addition the fixed costs f_j are negligible, a generalised assignment problem (GAP) (see Cattrysse and Van Wassenhove¹⁸) is obtained.

Alternatively, the problem can be stated by introducing variables w_{ijk} which express the amount transported from plant i via depot j to customer k . In this case, the following formulation results.

$$\min \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} q_{ijk} w_{ijk} + \sum_{j \in J} f_j y_j$$

subject to (2)–(5), (10), (11) and

$$\sum_{i \in I} w_{ijk} = d_k z_{kj} \quad \forall j \in J, k \in K,$$

$$\sum_{j \in J} \sum_{k \in K} w_{ijk} \leq p_i \quad \forall i \in I,$$

$$\sum_{k \in K} w_{ijk} \leq p_i y_j \quad \forall i \in I, j \in J,$$

$$w_{ijk} \geq 0 \quad \forall i \in I, k \in K, j \in J,$$

where $q_{ijk} = t_{ij} + c_{kj}/d_k$ is the cost per unit to supply customer k by plant i and depot j . An advantage of this formulation is that it allows us to model transportation costs q_{ijk} which depend on the plant-customer relation.¹¹ However, if these costs can be split into two linear parts for the transport from the plant to the depot and from the

depot to the customer, the first formulation is more suitable since it has far fewer variables. Furthermore, in contrast to two-level problems which also involve locational decisions on the first echelon,⁷ both formulations are also equivalent with respect to the LP relaxation: If $(\bar{w}, \bar{y}, \bar{z})$ is feasible for the LP relaxation of the second formulation, then $(\bar{v}, \bar{y}, \bar{z})$, where $\bar{v}_{ij} = \sum_k \bar{w}_{ijk}$, is feasible for the LP relaxation of the first formulation and has the same objective value. Conversely, if $(\bar{v}, \bar{y}, \bar{z})$ is feasible for the LP relaxation of the first formulation, then $(\bar{w}, \bar{y}, \bar{z})$, where $\bar{w}_{ijk} = \bar{v}_{ij} d_k \bar{z} / \sum_k d_k \bar{z}_{kj}$ if $\bar{y}_j > 0$ and $\bar{w}_{ijk} = 0$ otherwise, is feasible for the LP relaxation of the second formulation and has the same objective value.

Valid inequalities

The TSCFLP contains various subproblems (knapsack problems, the SPLP and the CFLP) for which valid inequalities, facets and separation procedures have been described in the literature. The inequalities used here to strengthen the LP relaxation of the TSCFLP are described briefly in the following.

Cover inequalities

Together with the integer requirements (10) and (11), the individual capacity constraint (3) and the aggregate capacity constraint (5) define knapsack polytopes for which lifted cover inequalities can be generated. Given a knapsack polytope

$$X^{KP} = \text{conv} \left(\left\{ x \in \{0, 1\}^n : \sum_{j=1}^n a_j x_j \leq b \right\} \right),$$

a minimal cover $C \subset N = \{1, \dots, n\}$ is a subset of the set N of all items such that $\sum_{j \in C} a_j > b$ and $\sum_{l \in C \setminus \{j\}} a_l \leq b$ for all $j \in C$. The corresponding inequality $\sum_{j \in C} x_j \leq |C| - 1$ is valid for X^{KP} and is called a cover inequality. If the cover C is partitioned into two disjoint subsets C_0 and C_1 , the inequality $\sum_{j \in C_0} x_j \leq |C_0| - 1$ is only valid for the KP if $x_j = 1$ for all $j \in C_1$. To obtain a valid inequality for the higher-dimensional polyhedron X^{KP} , the coefficients of the variables $j \in C_1$ have to be down-lifted. Furthermore, coefficients of variables $j \in N \setminus C$ may be up-lifted to strengthen the inequality. Let $L \subset N_0 \setminus C_0$ denote the set of variables whose coefficient has already been lifted, and $\sum_{j \in C_0 \cup L} g_j x_j \leq g_0 = |C_0| - 1 + \sum_{j \in C_1 \cap L} g_j$ the actual inequality, where $g_j = 1$ for $j \in C_0$. Define

$$\varepsilon(b_k) =$$

$$\max \left\{ \sum_{j \in C_0 \cup L} g_j x_j : \sum_{j \in C_0 \cup L} a_j x_j \leq b_k, x_j \in \{0, 1\} \forall j \in C_0 \cup L \right\}.$$

The lifting coefficient g_k of a variable $k \in (N \setminus C_0) \setminus L$ is then given by $g_k = g_0 - \varepsilon(b - \sum_{j \in C_1 \setminus L} a_j - a_k)$ if

$$k \in N \setminus C, \quad \text{and} \quad g_k = \varepsilon(b - \sum_{j \in C_1 \setminus L} a_j + a_k) - g_0 \quad \text{if} \quad k \in C_1.$$

For a given lifting sequence, the interrelated lifting problems can be solved by a dynamic programming approach which interchanges the roles of the constraint and objective function.^{19,20} An initial cover C and a lifting sequence can be obtained using a proposal of Gu *et al.*²¹ Gu *et al.* determine the initial cover C by including variables j according to nonincreasing values x_j^* of the actual fractional solution until a cover is reached, and then removing variables again until a minimal cover is obtained. Furthermore, they first up-lift the variables in $F = \{j \in N \setminus C : x_j^* > 0\}$ and finally up-lift the variables in $(N \setminus C) \setminus F$ in order of nonincreasing magnitude of reduced costs.

Odd hole inequalities

Odd hole inequalities are valid inequalities for the set packing problem (SPP)^{22–24} which is defined by

$$\max \{cx : Ax \leq 1, x_j \in \{0, 1\} \forall j\},$$

where A is a 0-1 matrix. By constructing the intersection graph $G = (V, E)$, which has a vertex for each variable x_j and an edge for each pair (x_i, x_j) of variables which must not be set both equal to one, the SPP can be transformed into a node packing problem (NPP). The NPP is to find a maximum cardinality subset $S \subseteq V$ of vertices such that no two vertices in S are connected. An odd hole is a chordless cycle of the graph G which is induced by a subset $O \subseteq V$ containing an odd number of at least 5 vertices. An odd hole gives the inequality

$$\sum_{j \in O} x_j \leq \frac{|O| - 1}{2}$$

which is valid for the NPP and the SPP.

The SPLP, which results from the TSCFLP if the constraint (3) and (5)–(9) are ignored, can be rewritten as an SPP by introducing the complemented variables $\bar{y}_j = 1 - y_j$. Odd hole inequalities for the SPLP have been considered by Cornuéjols and Thizy²⁵ and Cho *et al.*^{26,27} and shown to be special cases of a broader class of combinatorial inequalities for the SPLP. With respect to the SPLP, the intersection graph G has a special structure. In each odd hole of G , every third node corresponds to a location variable $\bar{y}_j$ while the two intermediate nodes correspond to assignment variables z_{kj_1} and z_{kj_2} belonging to the same customer k (Figure 1).

The edges of the odd hole correspond to constraints $z_{kj} + \bar{y}_j \leq 1$ and $z_{kj_1} + z_{kj_2} \leq 1$ respectively. If the edges are weighted with the slack of these constraints at the current fractional solution (y^*, z^*) , the corresponding odd hole inequality is violated, if the cycle has a weight less than 1. Therefore, violated odd hole inequalities can be found by computing minimum weight odd cycles of the intersection graph. As Caprara and Salazar²⁸ show for a generalisation

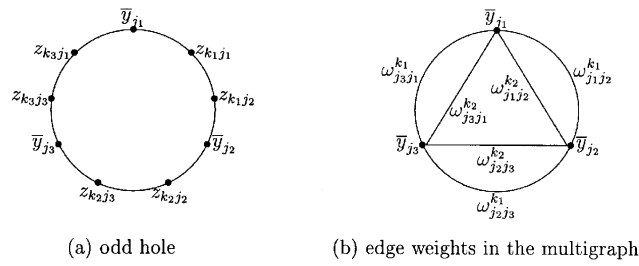


Figure 1 Odd holes in the SPLP.

of the SPLP, this procedure can be simplified considerably in the case of an SPLP: it suffices to compute minimum weight odd cycles in a multigraph which has a vertex for each fractional location variable $\bar{y}_j$ and at most $|K|$ edges of weight $\omega_{j_1j_2}^k = 3 - \bar{y}_{j_1}^* - \bar{y}_{j_2}^* - 2(z_{kj_1}^* + z_{kj_2}^*)$ for each pair $\bar{y}_{j_1}, \bar{y}_{j_2}$ of $\bar{y}$ -nodes (Figure 1). In the case of the TSCFLP, odd hole inequalities can be generated for both distribution stages using this procedure and the substructures defined by $\{(2), (4)\}$, and $\{(6), (8)\}$ respectively.

Flow cover and submodular inequalities

Aardal *et al*¹⁰ derive a general family of so-called submodular inequalities which give valid inequalities and facets for the CFLP. The CFLP results from the TSCFLP if the constraints (6)–(9) are omitted and the single-source requirements (1) are replaced by $z_{kj} \geq 0$. In order to derive the inequalities, Aardal *et al* looked at the maximum flow $f(J')$ which can be sent from a subset $J' \subset J$ of depots to a subset $K' \subset K$ of customers using a set of arcs $\{(j, k): j \in J', k \in K' \subseteq K\}$. Since $f(J')$ —as a function of the set $J' \subseteq J$ of depots—is submodular,¹⁰ the reduction in flow which results if some of the depots $j \in J'$ are closed is underestimated by the sum of the flow reductions $\rho_j(J') = f(J') - f(J' \setminus \{j\})$ which result if only one facility $j \in J'$ is closed. This argument shows the validity of the inequalities

$$\sum_{j \in J'} \left(\sum_{k \in K'_j} z_{kj} d_k \right) + \sum_{j \in J'} \rho_j(J') (1 - y_j) \leq f(J').$$

Aardal *et al* characterised special cases of these inequalities and give necessary and sufficient conditions for the resulting inequalities to be facet defining for the CFLP. The first class of inequalities results if $K'_j = K'$ for all $j \in J'$ and if J' is a cover of the set K' of customers (that is, $\lambda = \sum_{j \in J'} s_j - d(K') > 0$). In this case, $f(J') = d(K')$ and $\rho_j(J') = \max\{0, s_j - \lambda\}$ for all $j \in J'$. The inequality obtained is known as a flow cover inequality.^{19,29} In the second and third class of inequalities, the set J' of depots is partitioned into three disjoint subsets J'', Q and P . The depots $j \in P$ exclusively serve subsets $\bar{K}_j \subset K'_j$ of customers and may serve all customers $k \in K'' = \bigcup_{j \in J'' \cup Q} K'_j$ such that $s_j > d(K'_j)$ for all $j \in P$. Furthermore, $K'_j = K''$ for all

$J \in J'', K'_j \cap K'_{j_2} = \emptyset$ for all $j_1, j_2 \in Q$, and the depot set $J'' \cup Q$ covers the customer set K'' such that $s_j > d(K'_j) > \lambda$ for all $j \in Q$. In this case, $f(J') = d(K')$, $\rho_j(J') = \max\{0, s_j - \lambda\}$ for $j \in J'', \rho_j(J') = d(K'_j) - \lambda$ for $j \in Q$ and $\rho_j(J') = d(K'_j) > s_j - \lambda$ for $j \in P$. Aardal *et al* called the resulting inequality a ‘single depot inequality’ if $P \neq \emptyset$ and an ‘effective capacity inequality’ otherwise.

Regarding the TSCFLP, these inequalities can be derived for both distribution stages using the substructures defined by (2)–(4) on the one hand, and (6) and (8) and $\sum_{i \in I} v_{ij} \leq s_j y_j$ on the other. A separation heuristic given by Van Roy and Wolsey²⁹ can be used to find violated flow cover inequalities for a fixed customer set K' ; it is based on computing a minimal cover $J' \subset J$ of the customer set K' and checking the resulting flow cover inequality for violation. For the purposes of identifying violated submodular inequalities of the types described above, Aardal⁹ gives a separation heuristic: starting with the depot set J' , consisting of one depot with a fractional value, and the customer set K' , consisting of the customers served by this depot, the sets J' and K' are augmented until a cover is obtained; then, after checking the conditions above and altering the sets J' and K' if necessary, either a flow cover, effective capacity or single depot inequality is produced which is then checked for violation.

Computation of upper bounds

After optimising the current LP formulation, the information of the LP solution (v^*, y^*, z^*) can be used to construct a feasible solution. This task is divided into the following three steps:

- (1) Determination of the set of open depots
- (2) Determination of a feasible customer assignment
- (3) Solution of the transportation problem at the first distribution stage.

Determination of the set of open depots

In order to decide which depot should be open, four different methods are used.

The first two heuristics are simple rounding procedures. The first rounding method selects each depot whose LP value y_j^* exceeds 0.5. If the capacity of the open depots is not sufficient, further depots are opened in order of nonincreasing LP values. The second rounding approach first assigns each customer k to the depot j for which $y_j^* > 0.5$ and $z_{kj}^* > 0.5$ hold. The remaining unassigned customers k are assigned to the depot $j = \arg \max_j z_{kj}^*$; in the event of a tie, an assignment to a depot which has already been selected is preferred.

If the depot capacity constraint (3) and flow conservation constraint (7) are relaxed in a Lagrangian manner using multipliers $(\mu_j)_{j \in J}$ and $(\pi_j)_{j \in J}$ respectively, the resulting

subproblem can be reduced to an SPLP with assignment costs $\bar{c}_{kj} = c_{kj} + d_k(\pi_j + \mu_j)$. This SPLP can be solved heuristically by computing a ‘small’ subset S of the set of depots with positive LP value y_j^* such that a low assignment cost $\sum_k \min_{j \in S} \bar{c}_{kj}$ results and every customer can be covered by a depot $j \in S$ with $z_{kj}^* > 0$. This is achieved by performing the following steps:

- (1) Set $F = \{j \in J : y_j^* > 0\}$, $S = \{j \in F : |\{k \in K : z_{kj}^* = 1\}| \geq 1\}$ and $K_u = \{k \in K : \sum_{j \in S} z_{kj}^* = 0\}$.
- (2) As long as $K_u \neq \emptyset$ select a $k \in K_u$ and set $S := S \cup \{j_k\}$ and $K_u := K_u \setminus \{l \in K : z_{lj_k}^* > 0\}$, where $\bar{c}_{kj_k} = \min_{j \in F} \bar{c}_{kj}$.

This procedure closely resembles Erlenkotter’s⁴ primal heuristic for the SPLP. The current optimal dual variables of the constraints (3) and (7) are used as multipliers μ_j and π_j respectively. Added inequalities, other than the trivial cliques (4), are ignored when defining the SPLP, since the computational tests performed showed that no improvement in the solution quality is obtained if the added cuts are also relaxed in a Lagrangian manner.

The fourth procedure is based on the Lagrangian relaxation of the demand constraints (2), the flow conservation constraints (7) and all added inequalities $r \in R$ (except the trivial cliques (4))

$$\sum_{i \in I} \sum_{j \in J} \alpha_{ijr} v_{ij} + \sum_{k \in K} \sum_{j \in J} \beta_{kjr} z_{kj} + \sum_{j \in J} \gamma_{jr} y_j \leq \delta_r. \quad (12)$$

The current optimal dual variables, say $(\eta_k^*)_{k \in K}$, $(\pi_j^*)_{j \in J}$ and $(\theta_r^*)_{r \in R}$ for constraints (2), (7) and (12), are used as multipliers. The resulting Lagrangian subproblem is:

$$Z_{LR} = \min \sum_{j \in J} \left(\sum_{i \in I} \tilde{t}_{ij} v_{ij} + \sum_{k \in K} \tilde{c}_{kj} z_{kj} + \tilde{f}_j y_j \right) + \sum_{k \in K} \eta_k^* - \sum_{r \in R} \theta_r^* \delta_r \quad (13)$$

subject to (3)–(5), (6) and (9)–(11), where $\tilde{t}_{ij} = t_{ij} - \pi_j^* + \sum_{r \in R} \theta_r^* \alpha_{ijr}$, $\tilde{c}_{kj} = c_{kj} + d_k \pi_j^* + \sum_{r \in R} \theta_r^* \beta_{kjr} - \eta_k^*$, and $\tilde{f}_j = f_j + \sum_{r \in R} \theta_r^* \gamma_{jr}$. Since $v_{ij} = p_i$ if $\tilde{t}_{ij} = \min_{l \in J} \tilde{t}_{il} < 0$ and $v_{ij} = 0$ otherwise in an optimal solution of the program (13), the subproblem can be reduced to

$$Z_{LR} = \min \sum_{j \in J} \left(\sum_{k \in K} \tilde{c}_{kj} z_{kj} + \tilde{f}_j y_j \right) + \sum_{k \in K} \eta_k^* - \sum_{r \in R} \theta_r^* \delta_r - \sum_{i \in I} \lambda_i^* p_i \quad (14)$$

subject to (3)–(15) and (10)–(11), where the $\lambda_i^* = \max\{0, -\min_{j \in J} \tilde{t}_{ij}\}$ are the optimal dual variables of the constraints (6). The same subproblem arises if Lagrangian relaxation of the demand constraints (2) is applied to the CFLP with single sourcing. As Sridharan³⁰ shows, the

lower bound program (14) decomposes into the knapsack problems

$$Z_{LR} = \min \left\{ \sum_{j \in J} v_j y_j : (5) \text{ and } (11) \right\} + \sum_{k \in K} \eta_k^* - \sum_{r \in R} \theta_r^* \delta_r - \sum_{i \in I} \lambda_i^* p_i \quad (15)$$

where for all $j \in J$

$$v_j = \tilde{f}_j + \min \left\{ \sum_{k \in K} \tilde{c}_{kj} z_{kj} : \sum_{k \in K} d_k z_{kj} \leq s_j, z_{kj} \in \{0, 1\} \forall k \in K \right\}. \quad (16)$$

The lower bound Z_{LR} may improve the current LP bound. However, the improvement is generally negligible. Furthermore, the relaxation is only used here to construct a feasible solution from that of the current LP relaxation. In order to reduce the effort required to solve the above relaxation, variables x_{kj} and y_j which are integer-valued in the solution of the current LP relaxation are therefore fixed to their LP value.

Determination of a feasible customer assignment

Given a set S of open depots which covers the total demand, the task of finding a feasible customer assignment can be reduced to a GAP if the flow conservation constraints (7) and the added cuts are relaxed. While the constraints (7) are relaxed in a Lagrangian manner using the current optimal dual variables $(\pi_j^*)_{j \in J}$ as multipliers, the added cuts are ignored when defining the GAP

$$\min \left\{ \sum_{k \in K} \sum_{j \in S} c_{kj}^* z_{kj} : (2), (3) \text{ and } (10) \right\} \quad (17)$$

where $c_{kj}^* = c_{kj} + d_k \pi_j^*$.

Alternatively, the added cuts may also be relaxed in a Lagrangian manner using the corresponding dual variables as multipliers; however, the computational tests performed here did not show this procedure to offer any advantage in terms of solution quality.

The GAP (17) is solved approximately using simple reassignment heuristics. With the exception of the second rounding procedure, which directly yields a customer assignment, each customer k is initially assigned to the depot $j \in S$ which minimises $c_{kj} + (\pi_j^* + \mu_j^*) d_k$, where the μ_j^* are the current optimal dual variables of the depot capacity constraint (3).

Let $K_j \subset K$ denote the set of customers assigned to depot $j \in S$ and $V = \{j \in S : d(K_j) < s_j\}$ the set of depots whose capacity is violated. In order to obtain a feasible assignment, two different reassignment procedures are used. The first approach has been proposed by Barceló *et al*³¹ for the CFLP with single sourcing. The costs per unit demand of

reassigning a customer $k \in K_j$ from a depot $j \in V$ to a depot $h(k) \in S \setminus V$ are measured by

$$\rho_{h(k)} = \min_{l \in S \setminus V} \{(c_{kl}^* - c_{kj}^*)/d_k : s_l \geq d(K_l) + d_k\}.$$

Customers $k \in K$ who are assigned to a depot $j \in V$ are then reassigned in order of nondecreasing costs $\rho_{h(k)}$ to a depot $h(k)$ with sufficient capacity. The procedure stops if the solution is feasible or no reassignment can be performed; otherwise the reassignment costs are recomputed and the procedure is repeated. If no feasible solution is obtained by this procedure, the second reassignment approach tries to reduce the unfeasibility $\sum_{j \in S} \max\{0, d(K_j) - s_j\}$ of the solution by reassigning single customers or interchanging two customers between two depots. In each iteration this search procedure selects the ‘best’ neighboured solution in terms of the ratio of cost increase to decrease in infeasibility. Furthermore, if a feasible assignment has been reached by this means, the same local search method is used to reduce the assignment cost $\sum_{j \in S} \sum_{k \in K_j} c_{kj}^*$.

If a feasible customer assignment $\{K_j\}_{j \in S}$ has been determined by the above methods, the objective value of the solution found is computed by solving the transportation problem

$$\min \left\{ \sum_{i \in I} \sum_{j \in S} t_{ij} v_{ij} : (6), (7) \text{ and } (9) \right\}.$$

Finally, once the entire algorithm has been completed, a third heuristic is performed in order to further improve the customer assignment in the best solution found. To this end, the neighbourhood of the local search heuristic for the GAP is enlarged to allow changes which involve shifts and interchanges of up to two customers between pairs of open depots.

Using the reassignment procedures described above, a heuristic solution could be computed after each reoptimization step, that is the computation of the current LP solution, and for each of the four methods to find a set S of open depots. However, for large problem instances the resulting computational effort would be prohibitive. The computational effort is therefore reduced by the following means:

- A heuristic solution is only computed after each reoptimisation, which leads to a strict improvement of the lower bound.
- The reassignment steps are of course only performed for those of the four methods which differ in the initial customer assignment and yield a set S of depots with sufficient capacity.
- A partial solution given by the set S of open depots can only improve the objective value z_B of the best solution found if

$$\sum_{j \in S} v_j + \sum_{k \in K} \eta_k^* - \sum_{r \in R} \theta_r^* \delta_r - \sum_{i \in I} \lambda_i^* p_i < z_B \quad (18)$$

where the v_j are obtained from (16). The reassignment steps are therefore only performed if this condition is met. (In the implementation, the ‘dual depot values’ v_j are computed subject to the additional restriction that variables y_j and z_{kj} with an integral LP value are fixed to 0 and 1 respectively. It can therefore happen that a partial solution S is discarded even though it may lead to an improved overall solution.)

- The reassignment procedures are not allowed to remove customers k who are initially assigned to depot $j \in S$ with $z_{kj}^* \geq 0.9$.

Summary of the procedure

The main steps of the LP-based heuristic are summarised below. The sequence of cuts used in step 3 turned out to be favourable after some preliminary testing. Furthermore, submodular inequalities and odd hole inequalities may only be generated if no trivial cliques (4) and variable upper bounds (8) are violated.

Step 1 (Initialisation). Solve the weak LP relaxation which results from the LP relaxation of (1)–(11) by dropping constraints (4) and (8). If the LP solution satisfies the integer requirements, terminate. Otherwise go to step 2.

Step 2 (Computation of upper bound). Apply the four different methods for obtaining a set of open depots and an initial customer assignment. Apply the first and the second reassignment procedure to the different partial solutions (characterized by the initial customer assignment) obtained in this step which satisfy condition (18). If the assignment is feasible, solve the transportation problem and record the best current solution.

Step 3 (Cut generation). Identify and introduce into the problem formulation only violated inequalities of the same class in the following order

- (1) variable upper bounds (8) and trivial cliques (4),
- (2) lifted cover inequalities,
- (3) flow cover inequalities involving the total set of customers and plants respectively, using Van Roy and Wolsey’s²⁹ separation heuristic,
- (4) other submodular inequalities (flow cover, effective capacity and single depot inequalities) using Aardal’s⁹ separation heuristic,
- (5) odd hole inequalities

If such constraints have been found, go to step 4. Otherwise, go to step 5.

Step 4 (Reoptimizing the LP). Reoptimize the LP relaxation using the dual simplex method. If the LP solution satisfies the integer requirements, terminate. If the lower bound has strictly improved, go to step 2. Otherwise, go to step 3.

Step 5 (Improvement of customer assignment). Apply the third reassignment procedure to further improve the customer assignment in the best solution found.

Computational results

The LP-based heuristic was coded in Sun Pascal and run on a Sun Ultra (300 MHz) to solve test problems belonging to two different sets. To solve the LPs, CPLEX's³² callable library (version 4.0.8) was used.

The first set R contains 120 test problems which have been randomly generated in a similar way to the test problems of Cornuéjols *et al.*³³ for the CFLP. The set R contains 3 subsets of test problems which differ in the ratio $r = \sum_{j \in J} s_j / d(K)$ of total depot capacity to total demand. In each subset, there are 8 problems of each of the following sizes: $5 \times 10 \times 25$, $5 \times 25 \times 50$, $5 \times 25 \times 100$, $5 \times 50 \times 100$, $10 \times 50 \times 100$, $5 \times 25 \times 500$, $5 \times 50 \times 500$, $10 \times 50 \times 500$ where the first number is the number of plants, the second is the number of possible depot sites and the third the number of customers. Five instances of a test problem have been generated for each given size and ratio r . In the case of these test problems, which are available from 'www.unisg.ch/problems/logistic/problem.htm', the fixed depot costs and capacities reflect economies of scale.

For the test problems of the second set S , the shortest routes in a Swiss road network have been used to compute the transportation costs t_{ij} and the costs c_{kj} of assigning customers to depots, the set S contains 8 subsets. The test problems differ from one subset to another in the relative size of depot capacities, variable depot costs (as part of the assignment costs c_{kj}) and fixed depot costs, which have either been set to a relatively low or a relatively high value. Each subset again contains 1 problem instance of each of the following sizes $10 \times 25 \times 50$, $10 \times 25 \times 100$, $10 \times 50 \times 100$, $20 \times 25 \times 50$, $20 \times 25 \times 100$, $20 \times 50 \times 100$, $10 \times 50 \times 500$. For these problem instances, the ratio f_j/s_j of fixed depot cost to depot capacity is equal for the different potential depots sites $j \in J$.

The performance of the LP-based heuristic is assessed using some 'best known' lower and upper bounds. These bounds were computed either by CPLEX's procedure *mipoptimize()* or by a Lagrangian relaxation approach³⁴

which relaxes the capacity constraints (3) and (6). When using CLPPEX's MIP solver, a time limit of 4.5 hours was set for the test problems from set R , while no time limit was used for the problems from set S (except for the instances with 500 customers). The average gap between the best lower and upper bound is 0.25% for set R and 0.01% for set S .

The results obtained for the test problems from set R are summarised in Tables 2–4 (averages over the 5 instances of each test problem). In these tables, the number in the column headed *Size* denotes the size of the problem according to the 8 different problem sizes mentioned above, *LP* is the percentage improvement of the computed LP bound over the value of the strong LP relaxation, *Gap* the percentage deviation of the upper bound from the LP bound computed by means of the LP-based heuristic, *UB* the percentage deviation of the heuristic solution from the 'best known' lower bound, and *LB* the percentage deviation of the LP bound from the 'best known' upper bound. The terms *vub*, *lei*, *ohi*, *smi* denote the number of added variable upper bounds and trivial cliques, lifted cover inequalities, odd hole inequalities, and submodular inequalities as a percentage of the total number of constraints which are binding in the solution of the last LP. Furthermore, *It* is the number of LPs solved, *T_{LP}* the time spent to solve the LPs (including the time required to find violated inequalities), *T_H* the time spent to find feasible solutions, and *T_T* the total time in seconds of cpu time.

In a similar manner, Table 5 gives the results obtained for the test problems from set S . In order to save space, only averages are shown. In this table, the contraction *cvf* indicates if, for the corresponding subset of problems, the depot capacities, variable depot costs and fixed depot costs refer to a relatively large (*l*) or relatively small (*s*) value. The problem size is denoted by the letters *a*, ..., *g* corresponding to the 7 problem sizes in the set S as mentioned above.

For the random problems, the proposed method produces feasible solutions which deviate from the optimal solution or the best lower bound by 0.74% on average (4.28% in the worst case), while the lower bound deviates by 0.80% on average (2.95% in the worst case) from the objective value of the best known solution. For the Swiss problems, these percentage deviations amount to 0.31% (worst case 1.88%)

Table 1 Average deviation of lower and upper bound from optimality

	Set R				Set S
	$r = 1.5$	$r = 3.0$	$r = 5.0$	Total	
<i>UB</i>	1.23 (4.28)	0.60 (3.32)	0.33 (2.01)	0.52 (2.01)	0.27 (1.88)
<i>LB</i>	1.19 (2.95)	0.88 (2.17)	0.83 (2.60)	0.98 (3.25)	0.98 (3.25)
<i>Opt</i>	6 out of 40	16 out of 40	28 out of 40	50 out of 120	49 out of 56

Worst cases are shown in brackets.

Opt denotes the number of problems for which an optimal solution is known.

Table 2 Results obtained for the random problems with $r = 1.5$

<i>Size</i>	<i>LP</i>	<i>Gap</i>	<i>UB</i>	<i>LB</i>	<i>vub</i>	<i>lci</i>	<i>ohi</i>	<i>smi</i>	<i>It</i>	T_{LP}	T_H	T_T
1	2.77	2.85	1.42	1.37	20.0	26.7	0.4	1.7	25.2	0.17	0.05	0.22
2	0.80	1.48	0.96	0.94	8.6	28.4	0.2	1.0	28.4	0.69	0.26	0.95
3	0.47	1.15	0.62	0.74	20.0	14.9	0.1	0.8	35.2	1.61	0.59	2.20
4	0.37	1.71	1.59	1.16	6.1	28.4	0.1	0.2	30.4	3.40	1.83	5.23
5	0.33	1.78	1.59	0.88	7.3	27.1	0.0	0.3	28.6	3.00	1.60	4.60
6	0.10	0.62	0.39	0.48	37.8	2.7	0.2	0.7	61.8	21.99	13.20	35.19
7	0.05	1.02	0.94	0.42	23.8	7.8	0.0	0.4	47.0	27.74	9.41	37.14
8	0.05	0.86	0.77	0.33	26.9	7.2	0.3	0.3	45.0	27.04	9.23	36.27
ave	0.62	1.43	1.03	0.79	18.8	17.8	0.2	0.7	37.7	10.70	4.52	15.22
min	0.02	0.39	0.00	0.20	4.4	1.7	0.0	0.0	15.0	0.10	0.02	0.12
max	5.05	6.28	4.28	2.95	40.8	32.0	1.9	4.3	110.0	51.45	23.17	74.62

Table 3 Results obtained for the random problems with $r = 3.0$

<i>Size</i>	<i>LP</i>	<i>Gap</i>	<i>UB</i>	<i>LB</i>	<i>vub</i>	<i>lci</i>	<i>ohi</i>	<i>smi</i>	<i>It</i>	T_{LP}	T_H	T_T
1	2.16	1.44	0.79	0.63	47.0	4.7	5.1	0.2	17.0	0.14	0.02	0.16
2	0.92	2.19	0.72	1.43	35.5	8.1	2.0	2.2	36.4	0.91	0.16	1.06
3	0.67	0.94	0.32	0.61	50.3	2.7	3.2	1.9	62.6	3.19	0.79	3.99
4	0.18	1.64	1.04	0.95	32.4	7.1	0.3	0.6	32.4	2.92	0.58	3.49
5	0.18	1.61	0.90	1.02	33.5	5.6	1.1	1.1	34.4	2.85	0.56	3.42
6	0.35	0.63	0.41	0.42	57.2	0.5	1.3	1.1	84.4	36.50	36.08	72.58
7	0.10	1.07	0.89	0.61	56.1	0.8	2.1	0.6	99.8	76.28	54.10	130.38
8	0.18	1.21	0.97	0.65	55.1	0.6	2.1	1.1	106.6	108.95	70.23	179.19
ave	0.59	1.34	0.75	0.79	45.9	3.8	2.1	1.1	59.2	28.97	20.32	49.28
min	0.00	0.00	0.00	0.00	27.7	0.0	0.0	0.0	6.0	0.06	0.00	0.06
max	5.24	4.64	3.32	2.17	61.0	10.0	13.0	3.3	220.0	290.10	106.01	396.11

Table 4 Results obtained for the random problems with $r = 5.0$

<i>Size</i>	<i>LP</i>	<i>Gap</i>	<i>UB</i>	<i>LB</i>	<i>vub</i>	<i>lci</i>	<i>ohi</i>	<i>smi</i>	<i>It</i>	T_{LP}	T_H	T_T
1	2.03	0.66	0.09	0.56	61.3	3.7	1.1	0.5	16.8	0.16	0.02	0.18
2	1.07	2.62	0.72	1.85	55.4	1.4	4.9	1.2	43.8	1.36	0.23	1.59
3	0.59	1.11	0.16	0.93	63.7	0.6	2.7	2.3	52.8	3.67	0.82	4.49
4	0.24	1.16	0.29	0.98	49.7	2.3	3.9	1.4	59.6	5.88	1.13	7.01
5	0.42	1.31	0.57	0.73	53.7	2.1	4.5	1.2	59.6	6.68	1.19	7.87
6	0.18	0.40	0.26	0.15	63.0	0.6	0.2	1.0	78.4	30.22	9.57	39.79
7	0.11	0.77	0.57	0.47	67.3	0.5	1.7	0.7	113.2	98.45	56.13	154.58
8	0.17	1.15	0.85	0.79	65.9	0.4	2.2	1.6	166.4	180.79	124.51	305.31
ave	0.60	1.15	0.44	0.81	60.0	1.4	2.7	1.2	73.8	40.90	24.20	65.10
min	0.01	0.00	0.00	0.00	48.2	0.0	0.00	0.0	5.0	0.07	0.00	0.07
max	4.42	3.63	2.01	2.60	69.5	7.2	9.9	6.5	390.0	491.91	404.17	896.08

for the upper bound, and 0.97% (worst case 3.25%) for the lower bound. If only those problems for which an optimal solution is known are considered, we obtain the solution gaps given in Table 1.

By comparing the columns *UB* of Tables 2–4, it is apparent that the procedure tends to produce better solutions if the (depot) capacities are less restrictive (see also Table 1 and 5). This might be due to an increased difficulty of finding good customer assignments for small values of r

(which is also a major reason why such problems are hard to optimise). In contrast, the sharpness of the lower bound (see column *LB* in Tables 2–5) is not influenced by the strength of the capacity constraints. Furthermore, neither the quality of the upper bound nor the quality of the lower bound deteriorates with an increase in problem size.

The computation times are relatively low (on average 43 s for set R and 74 s for set S). On average, the largest problems (10 plants, 50 depot sites, 500 customers) required

Table 5 Results obtained for the Swiss problems

	<i>LP</i>	<i>Gap</i>	<i>UB</i>	<i>LB</i>	<i>vub</i>	<i>lci</i>	<i>ohi</i>	<i>smi</i>	<i>It</i>	T_{LP}	T_H	T_T
<i>Size</i>	<i>Averages for given problem size</i>											
a	1.73	0.90	0.14	0.76	52.6	3.4	1.7	1.7	29.9	0.96	0.19	1.15
b	1.67	1.33	0.17	1.13	56.1	1.9	7.3	1.7	82.5	5.85	2.91	8.76
c	0.60	1.66	0.46	1.18	43.2	4.5	2.9	2.1	66.8	6.16	1.83	7.99
d	2.18	0.74	0.12	0.61	58.7	3.5	1.6	1.6	42.5	1.68	0.36	2.04
e	1.79	1.90	0.20	1.66	58.4	1.6	5.0	1.7	64.5	4.35	2.10	6.45
f	0.41	1.89	0.81	1.08	44.9	5.6	1.7	2.3	63.3	5.99	2.14	8.14
g	0.93	0.56	0.24	0.37	54.2	1.4	3.2	0.5	177.6	170.85	313.84	484.69
<i>cnf</i>	<i>Averages for given problem characteristics</i>											
lll	2.40	1.33	0.10	1.20	60.2	1.8	4.1	1.6	101.9	36.86	5.37	42.23
lls	2.72	1.13	0.16	0.95	60.2	1.8	2.4	0.6	56.1	16.77	3.88	20.66
lsl	0.74	1.29	0.40	0.88	62.3	1.3	4.7	0.7	77.3	27.05	4.84	31.89
lss	1.37	1.62	0.41	1.19	58.4	1.8	5.1	1.5	108.3	92.52	246.00	338.51
sll	0.78	1.14	0.45	0.68	43.4	5.3	1.7	3.4	79.3	16.72	10.19	26.91
sls	1.38	1.16	0.17	1.00	40.6	4.5	4.6	1.8	58.6	7.75	3.70	11.45
ssl	0.46	1.05	0.44	0.64	48.2	3.8	3.2	2.1	56.4	10.03	88.10	98.13
sss	0.80	1.55	0.33	1.21	47.6	4.6	0.9	1.6	64.4	16.10	7.49	23.60
	<i>Total average</i>											
ave	1.33	1.28	0.31	0.97	52.6	3.1	3.3	1.7	75.3	27.98	46.20	64.17
min	0.12	0.00	0.00	0.00	29.7	0.2	0.0	0.0	15.0	0.52	0.04	0.59
max	5.52	3.51	1.88	3.25	73.0	13.6	20.6	10.3	400.0	617.86	1711.33	2329.18

about 3 min (15 min in the worst case) for set R and 8 min (40 min in the worst case) for set S (see column T_T in Tables 2–5). In comparison with the Lagrangian heuristic³⁴ mentioned above, which produces bounds in the magnitude of 0.1% (upper bound) and 0.5% (lower bound) from optimality, the LP-based heuristic is on average 12 times faster. The computation times required for the LP-based heuristic increase rapidly with problem size, especially in the case of a higher number of customers. At the same time, the proportion T_H/T_T of time devoted to the task of finding feasible solutions grows with the number of customers due to the increased effort required for reassigning customers (the third reassignment procedure in particular, which is called only once, can be time-consuming). However, the increase in computation time and the proportion of time required to perform the heuristics could be successfully limited by temporarily fixing customer assignments in cases where the corresponding variable has an LP value near to unity. (Without adopting this method, the computational effort was sometimes more than 10 times larger while on the whole the solutions obtained were no better).

As can be seen from the column headed LP in Tables 2–5, the relative improvement in the LP bound obtained by adding cuts of the types described above, declines with increasing problem size. As might be expected, the relative portion (vub) of active variable upper bounds and trivial cliques increases if the capacity constraints becomes less restrictive, while the proportion (lci) of generated lifted cover inequalities decreases and the proportion (ohi) of added odd hole inequalities again increases. Since variable

upper bounds and trivial cliques account for the largest proportion of added inequalities, more of these constraints need to be added and more reoptimisation steps performed if the relative depot capacities (or plant capacities) grow, resulting in higher computation times for these types of problems. (The increase in the value of r from $r = 1.5$ in Table 2 to $r = 5.0$ in Table 4 has doubled the number It of iterations and quadrupled the time T_{LP} required to solve the LPs; this behavior also becomes apparent from Table 5 by comparing the results obtained in the case of large capacities ($c = l$) with those obtained in the case of small capacities ($c = s$)). The relative strength of the different cuts cannot be concluded from the results shown in the tables. However, the general observations made during the computations have shown that lifted cover inequalities and submodular inequalities contributed most to an improvement in the LP bound, while the use of odd hole inequalities generally led to only very small improvements. On the other hand, the addition of odd hole inequalities in one step has often resulted in the detection of other types of cuts in the subsequent step.

Conclusions

In this paper, a heuristic solution procedure is proposed for a two-stage capacitated facility location problem (TSCFLP) with single source constraints. The approach is based on linear programming, and iteratively refines the LP formulation using valid inequalities and facets for various relaxations of the problem. After each reoptimisation step, a feasible solution is obtained from the current fractional

solution using different heuristics to determine the set of open depots and simple reassignment procedures to find a feasible customer assignment. The computational effort is reduced substantially without loss of solution quality by using dual information to check if a set of open depots looks promising and by using the primal LP solution to temporarily fix some customer assignments.

The procedure is tested on a large set of test problems with up to 10 plants, 50 potential depot sites and 500 customers. The computational results show that this method is able to compute near-optimal solutions and useful lower bounds for the TSCFLP in short computation times, even in the case of larger problem instances. On the other hand of course, the approach may become inefficient in the case of problems with a very large number of customers and relatively weak capacity constraints, since too many variable upper bounds and trivial cliques have to be added to solve the LP.

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